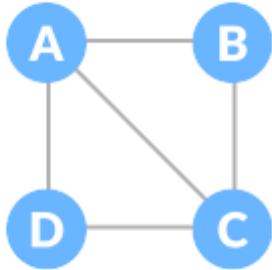


Spanning Tree and Minimum Spanning Tree

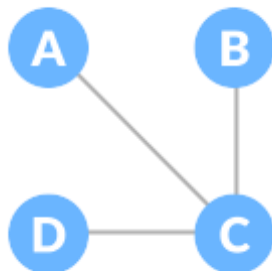
Before we learn about spanning trees, we need to understand two graphs: undirected graphs and connected graphs.

An **undirected graph** is a graph in which the edges do not point in any direction (i.e., the edges are bidirectional).



Undirected Graph

A **connected graph** is a graph in which there is always a path from a vertex to any other vertex.



Connected Graph

Spanning tree

A spanning tree is a sub-graph of an undirected connected graph, which includes all the vertices of the graph with a minimum possible number of edges. If a vertex is missed, then it is not a spanning tree.

The edges may or may not have weights assigned to them.

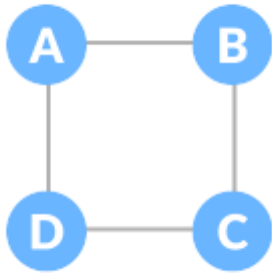
The total number of spanning trees with n vertices that can be created from a complete graph is equal to $n^{(n-2)}$.

If we have $n = 4$, the maximum number of possible spanning trees is equal to $4^{4-2} = 16$. Thus, 16 spanning trees can be formed from a complete graph with 4 vertices.

Example of a Spanning Tree

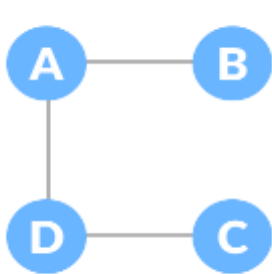
Let's understand the spanning tree with examples below:

Let the original graph be:

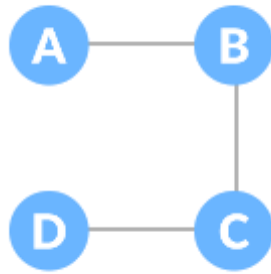


Normal graph

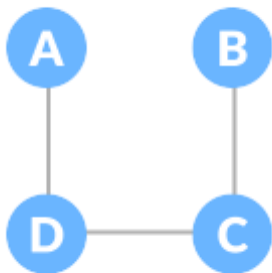
Some of the possible spanning trees that can be created from the above graph are:



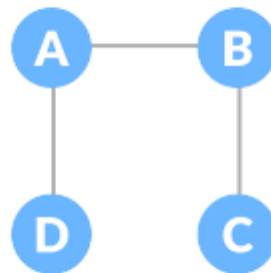
A Spanning Tree



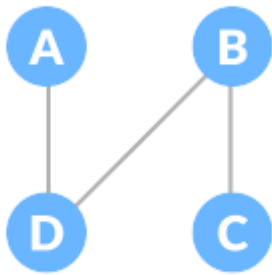
A Spanning Tree



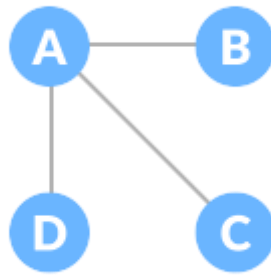
A Spanning Tree



A Spanning Tree



A Spanning Tree



A Spanning Tree

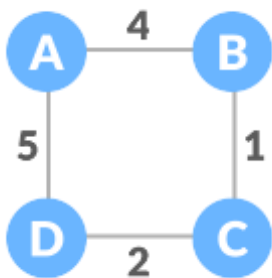
Minimum Spanning Tree

A minimum spanning tree is a spanning tree in which the sum of the weight of the edges is as minimum as possible.

Example of a Spanning Tree

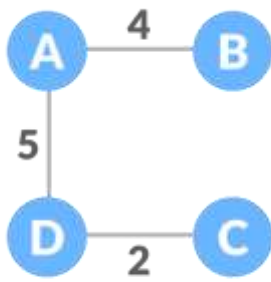
Let's understand the above definition with the help of the example below.

The initial graph is:

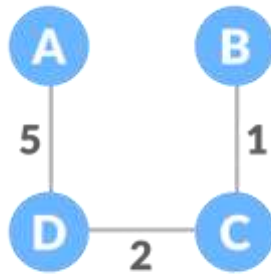


Weighted graph

The possible spanning trees from the above graph are:



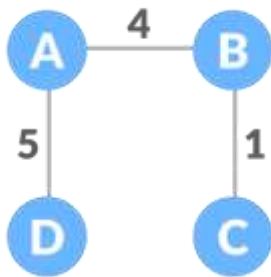
sum = 11



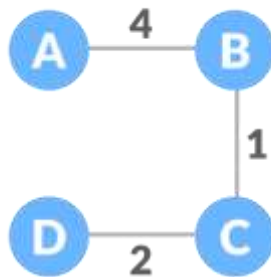
sum = 8

Minimum Spanning Tree - 1

Minimum Spanning Tree - 2



sum = 10

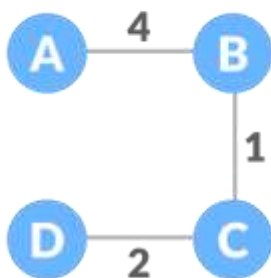


sum = 7

Minimum Spanning Tree - 3

Minimum Spanning Tree - 4

The minimum spanning tree from the above spanning trees is:



sum = 7

Minimum Spanning Tree

The minimum spanning tree from a graph is found using the following algorithms:

1. [Prim's Algorithm](#)
2. [Kruskal's Algorithm](#)

Spanning Tree Applications

- Computer Network Routing Protocol
- Cluster Analysis
- Civil Network Planning

Minimum Spanning Tree Applications

- To find paths in the map
- To design networks like telecommunication networks, water supply networks, and electrical grids.